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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

MAKE UP EXAMINATIONS – OCTOBER 2023

Program	:	**B.E. - Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Microcontrollers**	**Max. Marks**	:	**100**
Course Code	:	**IS44**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	List and analyze the features of RISC processors that broke with traditional design techniques used in CISC based processors.	CO1	(08)
	b)	List and explain the three types of timers in a microcontroller unit.	CO1	(07)
	c)	With a neat diagram, list the internal components of a typical Micro Controller Unit (MCU).	CO1	(05)
2.	a)	Which are the two modes of data transfer between MCU and peripherals? Explain the general functioning of interrupts.	CO1	(08)
	b)	List and explain three types of Reset in a typical MCU.	CO1	(07)
	c)	R in RISC is somewhat antiquated. Justify the statement.	CO1	(05)

UNIT - II

3.	a)	Discuss with appropriate diagram that illustrates the use of DMA controllers on most advanced MCUs.	CO2	(08)
	b)	Discuss on the use of memory banks for a 16- bit & 32 bit processor.	CO2	(06)
	c)	For the following operations, find out which flags will be set, (if any) for a 16-bit processor? i. Subtract 0xFE45 from 0x0978 ii. Compare A and B where A= 0x9987 and B= 0x9978.	CO2	(06)
4.	a)	Illustrate with appropriate diagram for the working principle of asynchronous read and write operation applied over a specific task for an SRAM chip type.	CO2	(08)
	b)	Discuss the various flags in embedded systems that indicates the results of each operation performed over a specific tasks.	CO2	(06)
	c)	Discuss the Code editor, Cross compiler, Debugger and Graphical user interface for an application program from the context of integrated programming environment.	CO2	(06)

UNIT - III

5.	a)	Write a block diagram of a typical ARM7 Processor. Explain model operation used in ARM7 processor.	CO3	(07)
	b)	Discuss the register set available in ARM architecture with necessary diagram.	CO3	(08)
	c)	List any ten conditional codes with suffix, flags and meaning in assembly programming of ARM7.	CO3	(05)

6.	a)	List and explain 8 interrupts in the vector table of the processor?	CO3	(08)
	b)	With a neat diagram, define components of AMBA bus.	CO3	(07)
	c)	List the features of ARMv4T instruction set architecture?	CO3	(05)

UNIT- IV

7.	a)	Write an ARM C program to display the string 'WEL COME TO ISE' in the serial window of UART1.	CO4	(10)
	b)	Briefly discuss the internal BUS structure of the LPC 214x family with appropriate diagram.	CO4	(10)
8.	a)	Write any SoC device architecture with its working principle and discuss how it is suitable for specific application.	CO4	(10)
	b)	Write an ALP to find the average of 10 numbers stored in memory, such that the numbers are stored in Read Only memory in address starting from "SOURCE".	CO4	(10)

UNIT - V

9.	a)	Explain the memory model of Cortex-MO and explain types and attributes of memory.	CO5	(08)
	b)	How does NVIC take care of nesting of interrupts?	CO5	(07)
	c)	List general purpose registers and other special purpose registers of Cortex-MO core.	CO5	(05)
10.	a)	Discuss the advantages and architectural features of Cortex-MO+.	CO5	(10)
	b)	List and Explain the interrupts of Cortex-MO and different states that each interrupts be in any time.	CO5	(10)

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